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From Molecules to Devices: A Multiscale Approach to Evaluating Organic Photovoltaics

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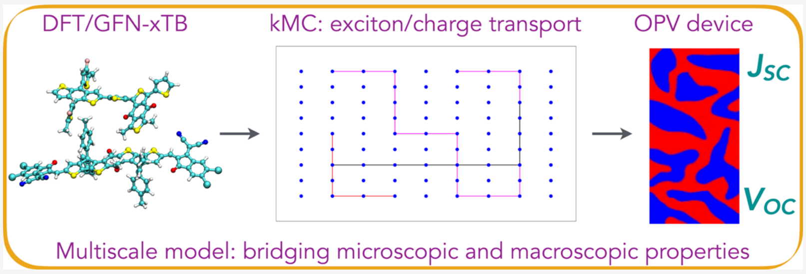

ABSTRACT: Due to their efficient molecular design, nonfullerene acceptors (NFAs) have significantly advanced organic photovoltaics (OPVs). However, the lack of models to screen and evaluate candidate NFAs based on the resulting device performance has impeded the rapid development of high-performance molecules. This work introduces a computational framework utilizing a kinetic Monte Carlo (kMC) model to derive device parameters from molecular properties computed through first principles. By analyzing the quantum chemical properties of diverse dimeric conformers, we estimate the relative probabilities of microscopic processes such as charge separation, recombination, and transport along with charge transfer state formation in the active layer of OPVs. These probabilities set up a random walk of charge carriers in a grid with disordered molecular sites, allowing us to track their average behavior and calculate key device parameters. Our model consistently predicts measured device parameters, including the short-circuit current and open-circuit voltage, for OPVs with diverse NFAs with high accuracy. Additionally, we applied the model to evaluate donor−acceptor combinations of known compounds and newly designed NFA molecules, identifying high-performing pairs. This model offers a computationally efficient approach for designing novel NFA molecules and optimizing the OPV performance.

1. INTRODUCTION

Over the past few decades, the efficiency of organic photovoltaics (OPVs) has seen significant improvements.[1] Central to these advancements is the bulk heterojunction (BHJ) configuration, where donor and acceptor domains are dispersed throughout a composite thin film.[2,3] This arrangement has been key to achieving high efficiencies. In BHJ devices, donor polymers (molecules) absorb photons to form excitons, which migrate to the donor−acceptor interface. The difference in energy levels of the donor and acceptor may drive the excitons to form a charge transfer (CT) state at the interface, whereby the electron is located in the acceptor molecule, while the hole remains on the donor molecule. If these are further separated into free electrons and holes, the charge carriers independently diffuse through the acceptor and donor domains, reaching the electrodes and generating current.[4,5] Recent research efforts have focused on the molecular design of donor and acceptor materials to enhance charge separation and transport efficiency. Notably, the development of nonfullerene acceptors (NFAs) has led to significant improvements in BHJ OPV device performance.[6−9] NFAs have not only improved the power conversion efficiency (PCE) but also enhanced the stability and tunability of OPVs, paving the way for their broader application and commercialization.[8,10] The greater impact of NFAs underscores the potential for further innovations in organic solar cell technology, driving the field toward more efficient and sustainable energy solutions.

Molecular design has significantly enhanced the performance of NFAs; however, the absence of a comprehensive model capable of predicting device properties based on molecular structure has hindered rapid screening and development. First-principles computations can accurately calculate molecular properties such as exciton binding energy, orbital energy differences, and quadruple moments.[11−13] While these properties are crucial, they are insufficient as standalone parameters for predicting NFA performance in OPV devices. The

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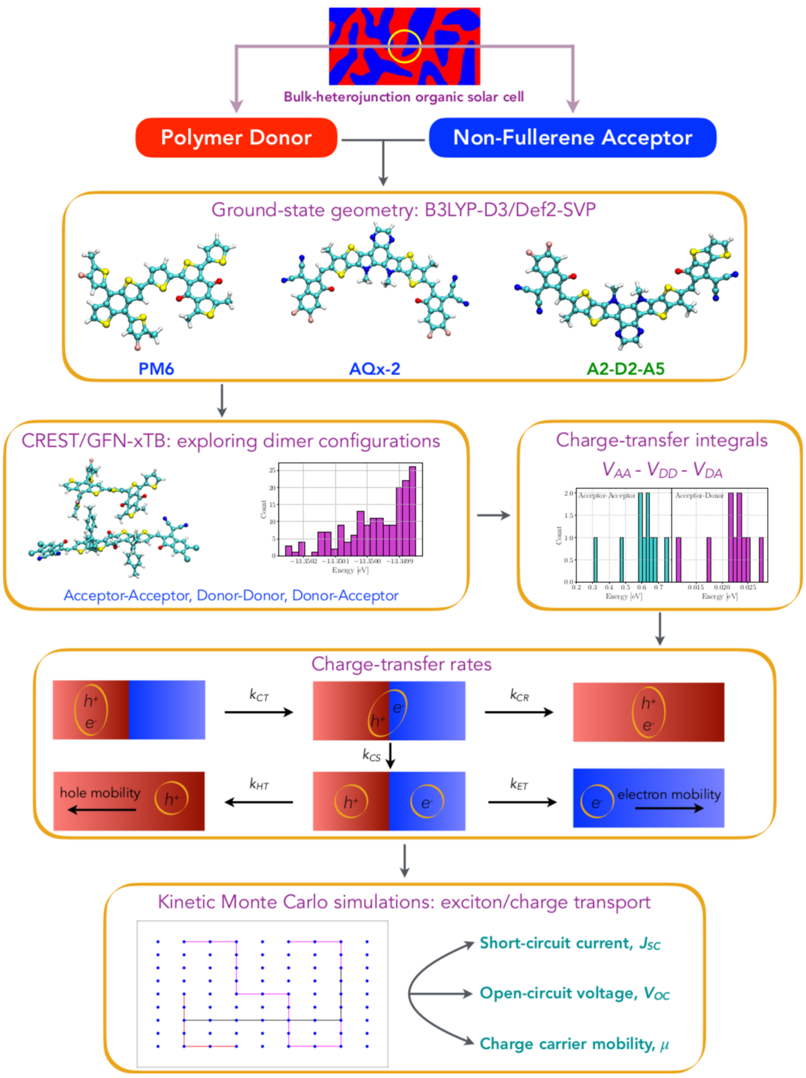

Figure 1. Schematic illustrating the process flow of our multiscale model. Initially, first-principles computations are performed to obtain relevant properties of the NFA molecules and the donor PM6. Configurations of dimers are then generated for both NFA−NFA and NFA−PM6 combinations, with quantities, such as electronic coupling, being computed for each configuration. The random walk of charge carriers is simulated on a grid of molecular sites, each possessing properties derived from first-principles calculations. These simulations are subsequently used to calculate key device parameters, allowing for a comprehensive evaluation of the OPV's performance.

performance of an OPV device largely depends on the interactions at the donor−acceptor interface, which has been overlooked in many past simulations.[14,15] To estimate device properties accurately, it is essential to evaluate donor and acceptor materials in combination rather than independently. Further, it is essential to include the effects of the disorder of the amorphous organic materials.[16] Charge carriers undergo various processes within an OPV device, including diffusion, separation, and recombination, and the efficiencies of these processes collectively determine the device characteristics. Given the crucial role of interfacial interactions in OPV operation,[17] comprehensive models that predict OPV parameters—such as short-circuit current (J_{SC}) and open-circuit voltage (V_{OC})—by integrating phenomena across different length scales are essential. Such models, which have not been extensively explored, would enable the computational screening of molecular structures to identify the most efficient donor−acceptor combinations, thereby bypassing the time-consuming process of experimental measurements. This approach would accelerate the discovery and optimization of high-performance NFAs for OPV applications.

In this work, we develop a multiscale model that utilizes parameters calculated from first principles as inputs for a kinetic Monte Carlo (kMC) framework. Our approach simulates the properties of donor and acceptor molecules, dimers, and domains to replicate the operational conditions in an OPV device accurately. Here, an entire BHJ is not simulated by first-principles methods as have been done before, by employing molecular dynamics or Monte Carlo-based methods.[18] Various configurations of molecular pairs are generated to form donor−acceptor domains, thereby incorporating both positional and energetic disorders. We employed a bilayer planar junction model to study the charge transport and recombination dynamics at the donor−acceptor interface. This simplified morphology allows for a clearer examination of the key interfacial effects, such as positional and energetic disorder, critical for determining device performance. By focusing on a bilayer system, we can systematically investigate the influence of these factors on the J_{SC} and V_{OC} without adding complexity

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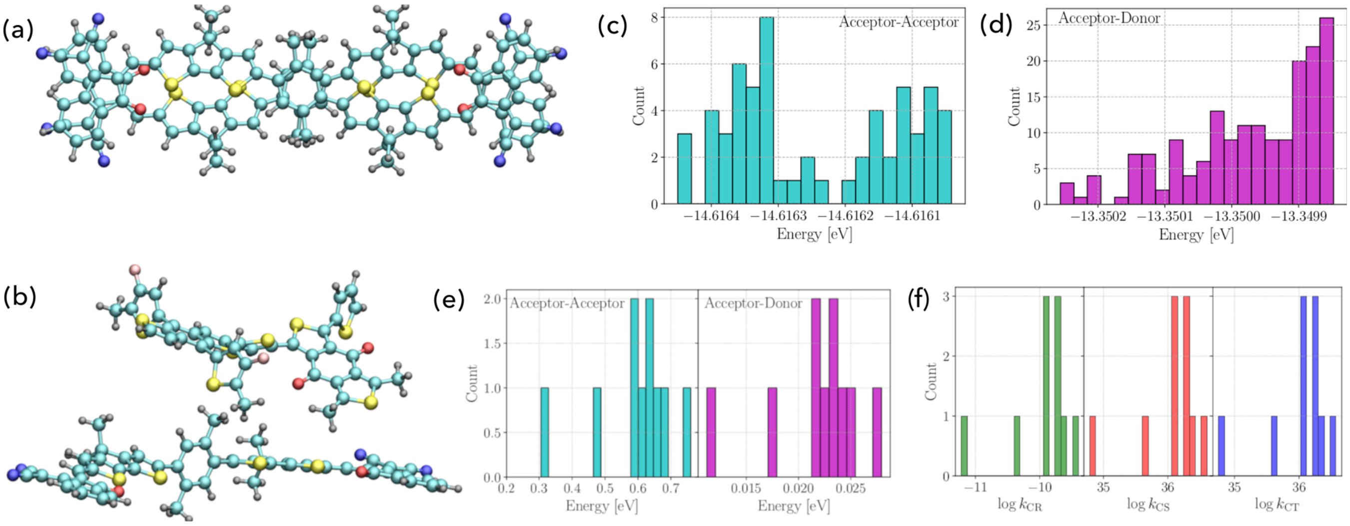

Figure 2. Representative molecular configurations obtained via CREST simulation: (a) AA dimer of DC6IC and (b) DA dimer of DC6IC-PM6. Histograms depicting the distribution of energy of various configurations of dimers obtained via CREST simulation: (c) AA dimer of DC6IC and (d) DA dimer of DC6IC-PM6. (e) Distribution of electronic coupling elements in the AA dimer of DC6IC and the DA dimer of DC6IC-PM6. (f) Histograms depicting distributions of various rates: k_{CT}, k_{CS}, and k_{CR} for various conformations of the PM6-DC6IC junction.

to BHJ architectures. Previously, kMC-based models were developed to study the charge transport behavior or device parameters of OPV devices. These models have been based on approximated values of various input parameters rather than values specific to the molecules under investigation.[19−23] On the other hand, models that have relied on calculations of molecular properties and generation of domain morphology tend to be computationally expensive.[24] We have developed a model that accounts for molecular properties, simulates key aspects of morphology, and includes processes undergone by charge carriers in OPV devices to predict device parameters with high accuracy. Unlike previous kMC models implemented for BHJ OPVs, our model emphasizes CT state formation at the donor−acceptor interface and its critical role in determining device performance.[23,25−27] We employ the Marcus relation to calculate the rates of various processes using quantities derived from ab initio methods, which enables us to determine the relative probabilities of these processes.[28] Our model successfully predicts crucial device properties, such as J_{SC} and V_{OC}, showing close agreement with experimental values. To the best of our knowledge, no kMC model has been demonstrated to predict the OPV parameters and evaluate various NFAs. We demonstrate that a multiscale model is essential for evaluating the performance of donor−acceptor combinations in OPVs. Our findings reveal that high-performing devices arise from specific donor−acceptor combinations that cannot be identified by evaluating individual molecules alone. Additionally, we employ our model to screen newly designed molecules, identifying promising candidates that, combined with the donor PM6, lead to high-performing devices.[29] The presented model can thus predict device parameters based on the molecular structures of donors and acceptors, facilitating the rapid screening and design of efficient OPV materials.

2. METHODS

We propose a multiscale methodology integrating first-principles simulations with a statistical kMC model. Initially, we calculated the properties of individual (isolated) molecules and dimers to simulate intermolecular CT. Subsequently, the kMC model simulates the trajectories of charge carriers within donor−acceptor domains, mimicking their behavior in OPV devices. The steps involved in our model are illustrated schematically in Figure 1. The detailed calculations and methodologies are described in the following sections.

2.1. Materials. Computations were performed on two groups of NFAs. The first group consisted of NFAs for which OPV devices have been previously tested (experimentally), while the second group comprised newly designed NFAs that have not yet been tested. In all simulations, PM6 was used as the donor, as shown in Figure S1. The first group of NFAs is depicted in Figure S2, in which OPVs have been fabricated and measured with donor PM6 for the molecules. For these molecules, computed values were directly compared to measured values to validate the model and its parameters. Molecules shown in Figure S3 have been used in OPVs with donors other than PM6, allowing us to compare the performance of the same NFA with those of different donors. Experimental details of these OPV devices are provided in refs 30−37. The details of donor−acceptor combinations studied in the experiment and also via kMC simulation are provided in Table S1. The second group of NFAs, whose structures are shown in Figure S4, was designed by linking known acceptor subunits of existing NFAs to known donor subunits to create new configurations, as reported in ref 29. Devices for these compounds have not been fabricated, but their molecular and dimer properties have been demonstrated to be optimal for OPV applications.

2.2. First-Principles Computations. Molecular geometries of all NFAs and the PM6 molecule were optimized within the density functional theory (DFT) framework, specifically at the B3LYP level of theory with the Def2SVP basis set and GD3 dispersion corrections.[38−41] The selection of the DFT functional and basis set was driven by our recent results, which showed this combination to be the most optimal for the study of NFAs.[29] Dimers of optimized NFA molecules adjacent to each other were generated to represent pairs found in the acceptor domains of the OPV devices. Similarly, dimers of PM6 molecules were created for the donor domain, and NFA−PM6 dimers were formed to simulate configurations at the donor−acceptor interface. To obtain various configurations within a thin film, statistical sampling of dimer geometries was performed using the conformer-rotamer ensemble sampling tool (CREST), employing the semiempirical tight-binding method GFN-xTB implemented in the xTB program.[42,43] These calculations were executed by using the CREST module of the `extended tight binding` (xTB) calculator. Examples of NFA−NFA (AA) and NFA−PM6 (DA) dimers thus obtained are shown in Figure 2 for the DC6IC acceptor. Various configurations predicted by CREST, in the vicinity of the equilibrium configuration, span a range of energies, as displayed in Figure 2. From these, the five configurations with the lowest energy and the five with the highest energy were selected for further calculations. For each domain of any molecule, such ten configurations are considered to be randomly distributed throughout the film.

For each of the selected dimer configurations, the values of the CT integrals were calculated. For the DA dimers, the CT integral (V_{DA}) was calculated as per eq 1.[44]

$$V_{DA} = \frac{\mu_t \Delta E}{\sqrt{\Delta\mu^2 + 4\mu_t^2}} \tag{1}$$

Here, μ_t is the transition dipole moment obtained from computations of each dimer in the excited state employing time-dependent DFT (TD-DFT)-based methods. ΔE is the vertical excitation energy for the dimer (S_0 to S_1). $\Delta\mu$ was calculated by performing TD-DFT-based computations on the dimer with the electric field (F) set to 10^{-4} a.u., along with computations with $F = 0$, applying eq 2.[44]

$$\Delta E_F = \Delta E_0 - \Delta\mu \cdot F - \frac{1}{2}\Delta\alpha F^2 \tag{2}$$

where $\Delta\alpha$ is the difference in polarizability of the dimer under an electric field and in its absence and ΔE_F and ΔE_0 are the vertical excitation energies under an electric field and in its absence, respectively. Computations with as well as those without fields were performed using the package ORCA,[45] which employed the B3LYP level of theory and the basis set Def2SVP with a GD3 dispersion.

The CT integral among the dimers generated was calculated for all NFAs and PM6, based on eq 3:[46,47]

$$V_{DD} = \frac{E_{HOMO} - E_{HOMO-1}}{2} \tag{3}$$

where E_{HOMO} refers to the energy of the highest occupied molecular orbital (HOMO) of PM6 and E_{HOMO-1} energy of the orbital lower than HOMO. Appropriate functions were included to orthogonalize basis sets to account for spatial asymmetry among the dimers in different configurations.[48] Orbital energies were calculated by Gaussian 09[49] for PM6 molecules and the dimers, and the CATNIP package[48,50] was used to determine the value of V_{DD}. The CT integral for all NFAs (V_{AA}) was calculated in a similar way, where the lowest unoccupied molecular orbital (LUMO) replaces the HOMO in eq 3.

2.3. CT Rates. The rate of formation of the CT state at the donor−acceptor interface (k_{CT}) was calculated for each DA, along with the rate of charge separation (k_{CS}) and the rate of recombination (k_{CR}). These rates were determined based on the free energy changes associated with each process: CT state formation (ΔG_{CT}), charge separation (ΔG_{CS}), and charge recombination (ΔG_{CR}). The relevant equations are defined as follows.

$$\Delta G_{CR} = E^{D}_{HOMO} - E^{A}_{LUMO} \tag{4}$$

$$\Delta G_{CS} = -\Delta G_{CR} - E^{D}_{O} \tag{5}$$

$$\Delta G_{CT} = \Delta G_{CS} - E^{A}_{b} \tag{6}$$

Here, E^{D}_{HOMO} refers to the energy of HOMO of PM6, while E^{A}_{LUMO} refers to the energy of LUMO of the respective NFA. E^{D}_{HOMO} and E^{A}_{LUMO} values are obtained by performing DFT-based calculations on their monomers. E^{D}_{O} is the optical gap of the PM6 donor molecule, and E^{A}_{b} is the exciton binding energy of the NFA; both are obtained by TD-DFT computations on the respective molecules. All TD-DFT calculations were performed using the ORCA package.[45]

Substituting the calculated values, the rates for the processes were obtained according to the Marcus relation as per eq 7.[28]

$$k = \frac{V_{DA}^2}{\hbar}\sqrt{\frac{\pi}{\lambda kT}}\exp\frac{-q(\Delta G - \lambda)^2}{4\lambda kT} \tag{7}$$

where the values of ΔG are ΔG_{CT}, ΔG_{CS}, and ΔG_{CR} when the values calculated are k_{CT}, k_{CS}, and k_{CR}, respectively. λ refers to the reorganization energy. Internal reorganization energy was assumed to be 0.1 eV for NFAs and PM6; deviations from these values are not expected to impact the resulting values significantly, as reported in ref 51. While calculating k_{CT}, k_{CS}, or k_{CR} for DA junctions, λ was the sum of the donor and acceptor reorganization energy, and an outer reorganization energy of 0.3 eV was added to simulate the effects of thin-film media at the heterointerface.[52]

Similarly, the rate of electron transfer between acceptor molecules (k_{ET}) was calculated as per eq 8. The Marcus equation is a special case of the Marcus−Levich−Jortner equation, which may be applied to systems with $\lambda > kT$.[53]

$$k_{ET} = \frac{V_{AA}^2}{\hbar}\sqrt{\frac{\pi}{\lambda kT}}\exp\frac{-q(\Delta G - \lambda)^2}{4\lambda kT} \tag{8}$$

Here, ΔG is the energy difference between the sites participating in electron transfer. λ is the reorganization energy of the respective monomer. The rate of hole transfer in PM6 dimers (k_{HT}) was calculated based on eq 8 with V_{DD} in place of V_{AA}. All rates computed depend on the molecular properties and intermolecular interactions of the NFAs/PM6 in the bulk and at the interfaces and, hence, are distinct for each representative system.

2.4. Multiscale Model. The rates of various processes thus calculated were combined in a kMC model to simulate the microscopic processes in an operational BHJ OPV. Key device parameters, such as charge carrier mobility (μ), J_{SC}, and V_{OC}, were obtained from the kMC model. Mobility of the NFA was estimated by setting up a 3-D grid with each site corresponding to an NFA molecule. The center of the grid was initially considered to be occupied by an electron. k_{ET} was calculated for each of the 26 nearest neighbors of a grid point. The orientation of the grid points is randomly selected to be among those obtained from the CREST computations. The corresponding value of V_{AA} was included for rate calculation. Site energies were drawn from a Gaussian distribution of width 0.1 eV,[54] and their difference was considered ΔG. As σ is a property of the thin film and is found to vary across and within devices, an approximate value applicable to small molecules is considered. The values of k_{ET} were inverted and normalized to generate the probability of hopping to the neighbor, thus ensuring that the probability of hopping to all possible neighbors increased to 1. A cumulative sum of p of all probabilities is assigned to each site. A random number rn is generated such that $0 < rn < 1$ and is compared with the value of p at each site. Subsequently, the site assigned with p, which is the lowest value greater than rn, is selected by the electron as its next location. The electron is thus considered to have hopped to the selected neighbor, and the process is repeated. The time taken for each step is recorded as the inverse of the rate of hopping to the selected neighbor. The sequence is run for a total time (t) of 60 ns, for which the electron's displacement (r) from the initial site was recorded. Electron mobility (μ_e) was then calculated from diffusivity (D_e) according to eq 9 based on the square of average displacement ($\langle r\rangle$) of 4000 trials.[55]

$$\mu_e = \frac{q}{kT}D_e = \frac{q}{kT}\frac{\langle r^2\rangle}{t} \tag{9}$$

Here, q is the elementary charge, 1.62×10^{-19} C; k is the Boltzmann constant, 1.38×10^{-23} J K^{-1}; and T is the temperature, assumed to be 300 K. Hole transport mobility (μ_h) and diffusivity (D_h) for donor PM6 were calculated in the same way, with k_{HT} determining the hole transport process.

For the simulation of BHJ, a three-dimensional grid of molecular sites was set up with two domains of donors and acceptors, such that the interface was a plane in the middle. We recognize that real-world OPV devices typically exhibit more complex morphologies, such as BHJs, in which donor and acceptor materials are intermixed. These architectures introduce additional effects such as surface roughness and domain intermixing, which can influence charge generation and recombination processes. While our bilayer model captures essential interface physics, future work will focus on incorporating the morphology-dependent effects of BHJs further to improve the accuracy and generalizability of our predictions. Nonetheless, the bilayer model serves as a crucial foundation for understanding the fundamental mechanisms governing OPV performance.[17] Neighboring sites were assumed to be composed of dimers with molecular configurations selected above. An initial donor molecule was randomly selected as the site of photon absorption, whereby an exciton was generated. The exciton was considered to diffuse with rates (k_{XT}) from donor site i to site j, calculated per the Miller−Abrahams equation.[27]

$$k_{XT} = w\left(\frac{a_0}{r_{ij}}\right)^6 \begin{cases} \exp\left(-q\dfrac{E_j - E_i}{kT}\right) & E_j > E_i \\ 1 & E_j < E_i \end{cases} \tag{10}$$

Here, w is related to the attempt to hop frequency and can be estimated by the formulation given in ref 23 as 10^{-9} s^{-1}. a_0 is the equilibrium intersite distance of 4 Å,[56] and r_{ij} is the intersite distance between i and j, which have respective energy values E_i and E_j, in units of eV. Site energy values were drawn from a randomly generated normal distribution with a width of 0.1 eV. The calculated rates are a way of evaluating the various physical properties, such as energy and CT integral, of the molecules and interfaces for their performance in OPV.

An exciton that undergoes a random walk within the donor domain for a distance larger than the exciton diffusion distance L_D of PM6, estimated as 7 nm,[57] is considered lost. If the exciton reaches a donor site at the interface, the probability of the formation of a CT state was considered alongside the probability of the exciton diffusing to any neighboring donor sites. The probability is calculated, and the selection is performed according to the kMC process described above. Here, the exciton may select the event of CT formation or hop to another neighboring donor site. If the exciton selects the event of CT state formation, relative probabilities between k_{CS} and k_{CR} determine whether the CT was split into an electron and a hole. In the event of recombination, the exciton was considered to be lost. In the event of charge separation, electrons and holes were considered to diffuse independently, according to the rates of electron transfer among NFA molecules or hole transfer among donor molecules. At every hop, the distance of separation between electrons and holes was computed, and if evaluated to be less than the Coulomb radius estimated as 1.6 nm,[58] the probability of the pair forming a CT at nearby sites at the interface was considered. The CT may further separate or recombine as per the respective probabilities. Electrons (N_e) and holes (N_h) that were collected at the end of the domain were considered to have diffused to the electrodes and, hence, contributed to the current. At the end of all trials, the number of charge carriers collected was normalized with respect to trials as well as volume, such that $n_e = \frac{N_e}{4000 \times V}$ and $n_h = \frac{N_h}{4000 \times V}$, where V refers to the volume of the simulated domain. In this way, n_e and n_h represent the charge carrier density contributing to the current generated by the cell. In the absence of an electric field, the J_{SC} was estimated according to the displacement of electrons (x_e) and holes (x_h) from the interface as per eq 11.[59]

$$J_{SC} = qD_e\frac{dn_e}{dx_e} + qD_h\frac{dn_h}{dx_h} \tag{11}$$

Further, V_{OC} was estimated for all BHJs based on quantities computed via DFT and kMC, as per eq 12.[60,61]

$$J_{00} = qN_{CT} \times \left(\frac{2\pi}{\hbar}V_{DA}^2\right)\sqrt{\frac{1}{4\pi\lambda kT}} \tag{12}$$

$$J_0 = J_{00}\exp\left(q\frac{E_{HOMO}^D - E_{LUMO}^A}{2kT}\right) \tag{13}$$

$$V_{OC} = \frac{q}{kT}\log\left(\frac{J_0}{J_{SC}} + 1\right) \tag{14}$$

Here, N_{CT} refers to the surface density of the interacting donor and acceptor pairs, calculated from the molecular density of the donor−acceptor interface. Values of E_{HOMO}^D and E_{LUMO}^A were obtained by DFT-based computations on the isolated molecules, as described above.

The kMC simulation was run for 4000 trials, with each carried out for 60 ns. Here, one exciton is considered to be generated per trial. In the case of donor materials with higher generation rates for the same light intensity, more than one exciton may be simulated or vice versa. The average distance traversed by excitons, electrons, and holes from all trials was considered for further analysis. The estimated quantities, including μ_e, μ_h, V_{OC}, and J_{SC}, are all reported as the average values of four sets of trials. The standard deviation was

Table 1. Values of μ_e ($cm^2\ V^{-1}\ s^{-1}$), J_{SC} ($mA\ cm^{-2}$), and V_{OC} (V) Obtained via kMC Simulation Compared against the Available Experimental Measurements[30−37a]

NFA	μ_e^{kMC}	μ_e^{exp}	J_{SC}^{kMC}	J_{SC}^{exp}	V_{OC}^{kMC}	V_{OC}^{exp}
known systems						
AQx-2	7.34×10^{-5}	2.89×10^{-4}	23.82	25.40	0.95	0.86
BTP	4.84×10^{-5}	2.70×10^{-4}	24.77	26.84	0.89	0.84
Y11	8.83×10^{-5}	6.20×10^{-4}	26.22	24.60	0.94	0.84
AQx-1	6.98×10^{-5}	3.72×10^{-4}	23.72	22.18	0.91	0.89
BDTN-BF	7.25×10^{-5}	2.86×10^{-4}	20.32	20.20	0.85	0.93
BO-4F	5.96×10^{-5}	1.88×10^{-4}	23.60	26.04	0.86	0.83
BO-4Cl	1.01×10^{-4}	5.53×10^{-4}	19.16	26.03	0.85	0.84
HF-PCIC	7.91×10^{-5}	6.43×10^{-4}	17.61	17.00	0.87	0.91
cross-systems						
BT-CIC	2.71×10^{-5}	2.10×10^{-4}	22.39	22.50	0.93	0.70
DOC2C6−2F	3.98×10^{-5}	5.31×10^{-5}	16.47	21.35	0.95	0.85
BT-IC	2.23×10^{-5}	1.20×10^{-5}	13.61	17.50	0.94	0.81
AQx-16	7.66×10^{-5}		21.27	23.90	0.89	0.93
AQx-18	1.05×10^{-4}		20.60	25.00	0.88	0.93
DC6IC	5.98×10^{-5}		26.38	11.19	1.05	0.99

[a]For known systems, values are measured and simulated for the OPV of given NFAs with donor PM6. For cross-systems, values are measured for OPV of a given NFA with different donors, while they are simulated for OPV with PM6.

calculated to estimate the error in the kMC process. The parameters used in the kMC simulations are listed in Table S2.

A common and computationally inexpensive practice to obtain V_{OC} is performed via eq 15.[62]

$$qV_{OC} = |E^{D}_{HOMO} - E^{A}_{LUMO}| - 0.3 \quad (15)$$

In addition, V_{OC} has also been reported in the literature through the difference of ionization energy (IE) of the polymer donor and electron affinity (EA) of NFA, according to eq 16.[63]

$$qV_{OC} = |IE^{D} - EA^{A}| - 0.3 \quad (16)$$

In order to compare the kMC simulated V_{OC} with these two methods, we also obtained the V_{OC} via eqs 15 and 16. EA and IE were computed by optimizing the anionic state of all NFAs and the cationic state of PM6 by the same methods as those applied to neutral molecules; the difference in potential energy of the charged system and the neutral system was considered to be EA or IE.

3. RESULTS

3.1. Dimer Conformers. Simulations were conducted on a three-dimensional grid of molecular sites, forming adjacent donor and acceptor domains. Sites were simulated to be occupied by molecules of random orientation so as to replicate the amorphous nature of the domains. Configurations of donor−donor and acceptor−acceptor pairs in bulk as well as donor−acceptor pairs at the interface were generated via CREST simulations. Figure 2a presents an example of an AA pair conformer for DC6IC, while Figure 2b shows a sampled DA pair of PM6 with DC6IC. Figure 2c displays the potential energy range for all sampled AA pair conformations (corresponding to Figure 2a), which falls between −14.6165 and −14.6160 eV, indicating that the sampled conformations are within 0.5 meV of the equilibrium optimized configuration. Similarly, Figure 2d shows the energy range for DA pairs of DC6IC with PM6, ranging from −13.3503 to −13.3498 eV. A similar range of energy was obtained for the DA and AA pairs of all NFAs. Among the sampled geometries, 10 configurations were selected for simulation in the grid: five with the lowest energy values and five with the highest. This selection introduced disorder in the grid's morphology, representing the extremities of the conformers generated, thus forming amorphous domains and interfaces. As the distribution of energies of the various configurations is not broad, configurations with extreme energy values are selected and dispersed throughout the domains randomly to represent the variations in position and orientation among the molecules.

The intermolecular electronic coupling was calculated for the ten selected configurations of DD, DA, and AA pairs. Figure 2e shows the range of electronic coupling values for the AA and DA pairs of DC6IC, varying from 0.3 to 0.8 eV for AA pairs and from 0.01 to 0.03 eV for DA pairs. The electronic coupling is significantly higher for homojunctions (AA pairs) than for heterojunctions (DA pairs). Similar calculations were performed for all of the NFA configurations and DD pairs of PM6.

Based on the values of V_{DA}, the rates for charge separation, charge recombination, and CT state formation were calculated. The electron transfer rate and hole transfer rate were determined from the values of V_{AA} and V_{DD}. Figure 2f shows the ranges of k_{CT}, k_{CS}, and k_{CR} for various conformations at the PM6-DC6IC junction. The highest k_{CT} value is 7.65×10^{15} s^{-1}, with corresponding k_{CS} and k_{CR} values of 7.92×10^{15} and 8.29×10^{-5} s^{-1}, respectively. Among all NFA-PM6 interfaces, the highest k_{CS} and k_{CR} values are 1.64×10^{16} and 6.54×10^{4} s^{-1}, while the lowest values are 3.85×10^{12} and 1.34×10^{-5} s^{-1}, respectively. These rates represent the relative probabilities of each process occurring at any stage of operation of the OPV device.

3.2. Model Validation. The μ_e for all systems was calculated as per the kMC methodology, simulating a grid of pure NFA. Similarly, a grid of pure PM6 was simulated to obtain the μ_h of the donor. The rate of CT between pairs of dimers was calculated by using the values of V_{AA} or V_{DD}, and random occurrences of pairs of different configurations were considered to constitute the grid. Random walks of electrons in the respective grid were tracked based on these rates. μ_e was obtained from eq 9, the values of which are summarized in Table 1. The highest error in values obtained from the standard deviation of μ_e of four sets of trials is 5.36×10^{-5} cm^2

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$V^{-1}\ s^{-1}$. As shown in Figure 3a, the predicted μ_e values for all systems are close to the reported values. The root-mean-square error (RMSE) between the experimental and computed values is $2.4 \times 10^{-4}\ cm^2\ V^{-1}\ s^{-1}$. The computed μ_h of PM6 was $7.15 \times 10^{-5}\ cm^2\ V^{-1}\ s^{-1}$, which was close to the measured value of $3.2 \times 10^{-4}\ cm^2\ V^{-1}\ s^{-1}$.[64] This validates the methodology as well as the energetic disorder, reorganization of energy values, and the positional disorder introduced in the grid in the form of conformer variations. The calculated electron mobility is lower than the experimental value, which we attribute to the limitations of our dimer sampling approach. While full-scale molecular dynamics simulations could provide more realistic morphologies, they significantly increase computational cost. Our method balances computational efficiency and accuracy using xTB-based dimer sampling, which captures key interfacial interactions. Though this approach may introduce discrepancies, it provides a reliable estimate with reduced computational expense compared with MD-based methods. To further validate our strategy to gather configurations from CREST simulations, we performed all-atom classical molecular dynamics simulations on a representative system, AQx-2. These simulations were performed utilizing the OPLS-AA force field.[65,66] The details of MD simulations can be found in ref 67. From the MD simulated morphologies, we extracted dimer configurations on which quantum mechanical calculations were performed, as described above, to obtain the electronic couplings. Combining these with the rest of the kMC parameters, we obtained an electron mobility of $2.18 \times 10^{-6}\ cm^2\ V^{-1}\ s^{-1}$, reasonably close to the value predicted with the dimers generated by CREST simulations. The difference observed in mobility (MD-kMC approach) can be attributed to the underlying potential used in MD simulation, which is generalized and can often provide configurations that are different from those obtained from semiempirical potentials. While employing a computationally inexpensive approach, we show that the proposed method is better than a workflow involving the simulation of morphology by MD.

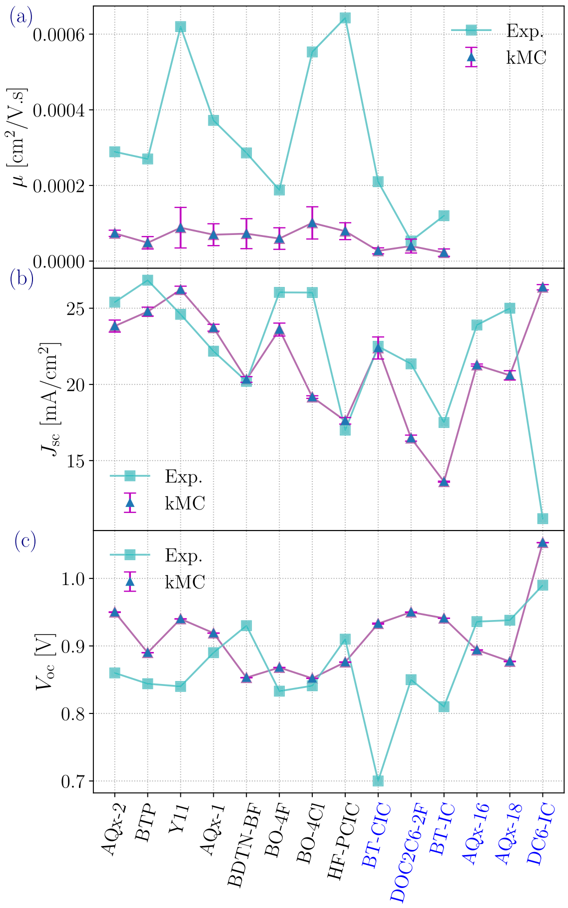

Figure 3. Comparison between kMC prediction and experimental measurements for (a) μ_e, (b) J_{SC}, and (c) V_{OC} for systems for which devices have been reported either with PM6 (labels in black text) or with another donor (labels in blue text). Reported values (experiments) are compared with those estimated by the multiscale model (kMC) and are found to be a close match.

The J_{SC} density was estimated for all devices by using eq 11, with the estimated values tabulated in Table 1. The maximum error in the computed values obtained via kMC as estimated from the standard deviation of four sets of trials is 0.25 mA cm^{-2}. The electron (n_e) and hole (n_h) densities were determined based on the random walk of charge carriers to the edge of the acceptor or donor domains. The number of lattice points constituting the simulation grid was set according to the intermolecular distance (d) in the direction of π-stacking, with larger intermolecular distances allowing for more lattice sites within a similar volume. The values of d ranged from 0.264 to 0.379 nm, consistent with previous reports on similar NFA molecules.[67] The predicted values of J_{SC} closely matched the experimental values, as shown in Figure 3b, thereby validating the kMC methodology. The RMSE between the experimental and computed J_{SC} considering all systems for which measured values are available with PM6 is 2.58 mA cm^{-2}. However, as can be seen from Figure 3b, the deviation of the computed value from the experiment is exceptionally high for the case of BO-4Cl. The competing rates of charge separation and recombination, as calculated, do not favor a high current for BO-4Cl. If this outlier is removed, then the RMSE is lowered to 1.50 mA cm^{-2}. This demonstrates that the predicted values are close to the experimentally reported values and that the general trends among the calculated values are consistent with those observed experimentally. Overall, these results validate the kMC approach for calculating J_{SC} as well as the specific algorithm implemented, demonstrating its effectiveness in evaluating and comparing OPV devices.

Among the NFA complexes where devices have been measured with a different donor polymer but simulations have been performed with the donor PM6, there are significant differences from measured device parameters. When paired with donor PM6, DC6IC leads to a high J_{SC} of 26.37 mA cm^{-2}, which is higher than the measured value of 11.19 mA cm^{-2} with the donor PBDB-T.[32] This observation indicates that the combination of DC6IC with PM6 results in a higher J_{SC} than with PBDB-T. In contrast, the combination of AQx-18 with PM6 results in a lower J_{SC} of 14.9 mA cm^{-2} compared to 25 mA cm^{-2} when paired with the donor D18.[36] To demonstrate the comparison of interfaces, we carried out simulations for the DC6IC system with PBDB-T, following the same protocol. The mobility of PBDB-T was found to be $2.43 \times 10^{-5}\ cm^2\ V^{-1}\ s^{-1}$, which is close to the experimental value of $1.16 \times 10^{-4}\ cm^2\ V^{-1}\ s^{-1}$.[68] Further, the calculated value of J_{SC} was found to be 13.07 mA cm^{-2}, and the value of V_{OC} was found to be 1.02 V. Calculated values are close to the measured values (see Table 1) and are lower than those estimated for the system of DC6IC with PM6, thereby confirming that the interface of DC6IC with PM6 leads to higher performing OPV than that of DC6IC

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and PBDB-T. It is apparent that OPV device parameters can be enhanced greatly by identifying high-performing donor−acceptor combinations. Besides the optimal properties of individual molecules, the properties of heterojunctions as well as homojunctions are crucial to OPV performance, which is included effectively in the kMC model. Hence, this method allows the simultaneous evaluation of individual molecules as well as donor−acceptor combinations in an efficient way that reduces significant experimental costs.

The V_{OC} was estimated for all BHJ devices, with the values provided in Table 1. The maximum error in the values of V_{OC} for all systems, as calculated from the standard deviation of four sets of trials, is 0.00083 V. As illustrated in Figure 3c, the estimated values are close to the experimental values for known systems. The RMSE of the difference between experimental and computed values among the NFAs for which experimental values with PM6 have been measured is 0.052 V. For the systems where devices have been measured with a different donor polymer, but simulations were performed with the donor PM6, V_{OC} values were also analyzed. Similar to the observations with J_{SC}, we find the OPV of AQx-18 to perform better with D18, with a V_{OC} of 0.93 V, rather than with PM6, with a V_{OC} of 0.88 V.[36] In contrast, the OPV of DC6IC exhibits a higher V_{OC} with PM6 at 1.05 V as compared to the OPV of DC6IC with PCE-10 at 0.99 V.[32] From the values of J_{SC} as well as V_{OC}, it is apparent that the OPVs of DC6IC and PM6 perform better than those of DC6IC with PCE-10, while the OPV of AQx-18 with D18 performs better than those of AQx-18 with PM6.

Previously, different methods of estimating the V_{OC} of an OPV have been considered. The V_{OC} for a specific donor−acceptor combination from first-principles can be obtained via eq 15.[62] However, other studies have suggested that eq 16 is a better estimate of V_{OC}.[63] We calculated the V_{OC} for the donor−acceptor pairs of OPV for the NFAs, for which experimental values with PM6 were measured by these methods. As shown in Table S3, the values of V_{OC} calculated by eq 15 or 16 are largely overestimated compared with the measured values. The RMSE for values of V_{OC} obtained by eq 15 is 0.161 V, while it is 0.707 V for values estimated by eq 16, which are significantly larger than the values estimated by eq 12. Values of V_{OC} estimated according to eq 12 depend on $E^{D}_{HOMO} - E^{A}_{LUMO}$ as well as on V_{DA}. Hence, eq 12 is a better predictor of V_{OC} by encompassing critical aspects of the donor−acceptor junction. This underscores the necessity of the kMC approach for evaluating OPV performance and highlights the effectiveness of the algorithm developed in this work.

3.3. Designed NFA Systems.

We implemented our model to evaluate the OPV performance with a set of newly designed NFAs, as presented in ref 29. The molecular structures, displayed in Figure S4, follow an acceptor−donor−acceptor architecture, specifically in the form of Ax-Dy-A5. A5 refers to a common acceptor subunit in all molecules, while Ax and Dy represent different acceptor or donor subunits, as per the nomenclature in ref 29. Based on their molecular and dimer properties, these structures were reported to be the highest performing among the ones studied in ref 29. Using our model, we performed an extensive evaluation of these molecules in OPV devices with a donor PM6. Figure 4 shows the J_{SC} and V_{OC} values obtained from the model for all of the molecules, with detailed values provided in Table 2. We will label the molecules as Ax-Dy, omitting the common A5 subunit for simplicity.

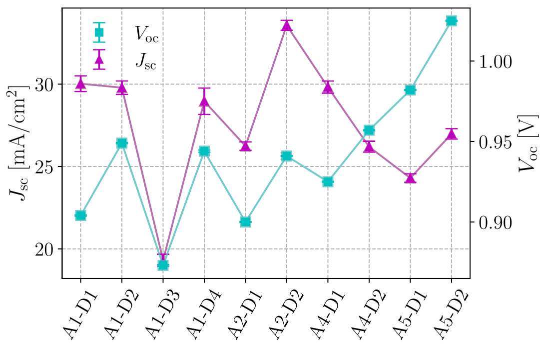

Figure 4. kMC simulated values of J_{SC} and V_{OC} for the newly designed compounds. A2-D2 and A1-D1 exhibit high values of J_{SC}, while A5-D1 and A5-D2 lead to high values of V_{OC}.

Table 2. Values of μ_e (cm^2 V^{-1} s^{-1}), J_{SC} (mA cm^{-2}), and V_{OC} (V) Obtained via kMC Simulations of the Designed Molecules[a]

NFA	μ_e	J_{SC}	V_{OC}
A1-D1	8.47×10^{-5}	30.02	0.90
A1-D2	5.59×10^{-5}	29.78	0.95
A1-D3	3.59×10^{-5}	19.30	0.87
A1-D4	6.95×10^{-5}	28.96	0.94
A2-D1	5.91×10^{-5}	26.24	0.90
A2-D2	3.00×10^{-5}	33.57	0.94
A4-D1	3.60×10^{-5}	29.81	0.92
A4-D2	5.63×10^{-5}	26.19	0.95
A5-D1	4.72×10^{-5}	24.30	0.98
A5-D2	4.46×10^{-5}	26.98	1.02

[a]PM6 was used as the donor polymer.

Our findings indicate high J_{SC} values for all molecules. The highest simulated J_{SC} is observed for the OPV of the molecule A2-D2 (33 mA cm^{-2}), followed by the OPV of A1-D1 (30 mA cm^{-2}). The V_{OC} values are also exceptionally high for the OPVs of A5-D1 and A5-D2, at 0.979 and 1.023 V, respectively. This is consistent with the reports of ref 29 that the symmetric molecules, i.e., with both acceptor subunits as A5, lead to high-performing OPVs, based on calculations of several molecular properties. However, OPVs of A5-D1 and A5-D2 exhibit lower J_{SC} values than those of A1-D1 and A2-D2. This highlights the necessity of performing kMC calculations to evaluate OPV device performance beyond individual molecular properties and to correctly analyze the molecules for their performance in OPV devices with a given donor.

4. DISCUSSION

The multiscale model consists of calculations of molecular and dimer properties by first principles, followed by the simulation of exciton and charge transport processes, CT state formation, separation, and recombination in a BHJ cell based on a kMC model. To include the effects of random molecular positions and orientations in a BHJ cell, multiple configurations of molecular dimers obtained by xTB/CREST simulations are considered within the bulk and interface, and the effect of configuration on intermolecular interactions is accounted for. As the values predicted by the model match the experimentally measured ones, the simulation of morphology as a dispersion of various configurations of dimers with a Gaussian distribution in energy values is valid. The results affirm the model's efficiency in replicating OPV devices and accurately predicting device characteristics.

J_{SC} and V_{OC} are critical indicators of device performance and are essential for determining the OPV efficiency. While various molecular properties have been identified as necessary for high-performance OPVs, such as high quadruple moment and optimized energy levels,[69,70] obtaining these properties independently is insufficient for evaluating OPV performance with a given NFA. This is particularly evident in the case of V_{OC}, where values estimated solely on the basis of first-principles computations (eqs 15 and 16) deviate significantly from the measured values as compared with those obtained here by combining computed quantities with considerations of configuration variations and the kMC methodology. Notably, the trends in J_{SC} or V_{OC} do not directly correlate with individual process rates. Instead, the OPV parameter values obtained from kMC simulations, which gauge the processes multiple times, closely match the experimental values. This observation underscores the necessity of performing kMC-based simulations to evaluate the OPV characteristics of donor−acceptor combinations, supplementing the electronic structure calculations. The multiscale model thus seamlessly connects the molecular structure to device operation. Further, it was demonstrated that the model is capable of identifying the most optimal donor−acceptor combinations for OPV performance. The evaluation of materials for OPV significantly reduces the need for tedious experiments to optimize the composition of BHJ thin films. A multiscale approach is necessary for a comprehensive evaluation of the NFA and, importantly, to identify the optimal donor−acceptor combination to maximize the OPV performance.

The values of electron mobility predicted by our model for NFAs are lower than the experimental values. The observed discrepancy between calculated and experimental electron mobility can be partly attributed to the limitations of our model in accounting for energetic disorder caused by charge−dipole interactions within the molecular solid. These interactions, arising from dipolar fluctuations, can create spatially varying energy landscapes that impact charge carrier mobility by inducing trap states or barriers. While our current approach models electron mobility based primarily on electronic coupling within dimer configurations, future enhancements could incorporate energetic disorder through either polarizable force fields in molecular dynamics simulations or Gaussian disorder models that include site energy variations due to dipolar effects.[71] These methods would enable more accurate predictions by explicitly reflecting the impact of local dipolar environments on charge transport.

In our study, we calculated the properties of the donor−acceptor interface directly from first-principles simulations. Although the bias potential at the interface (B), as outlined by Andrienko et al.,[72] is a critical factor in some models, we did not explicitly include it in our calculations. Instead, the computed values of CT rates inherently reflect the detailed molecular properties and interfacial interactions that arise from our donor−acceptor dimer simulations. While our dimer-based model derives CT rates from first-principles calculations that inherently capture molecular interactions—such as electrostatics, polarization, and dispersion—across donor−acceptor pairs, it does not explicitly incorporate the corrugated nature of the donor−acceptor interface, which can lead to a spatially dependent bias potential as discussed in ref 72. The bias potential at such an interface arises from local variations in the molecular orientation and separation across the interface, creating an additional layer of energetic heterogeneity beyond what is captured in a dimeric representation. Future extensions of our model could address this aspect by explicitly incorporating mesoscale structural features that represent interfacial corrugation and associated charge−dipole interactions, potentially enhancing the accuracy of the predicted charge dynamics.

The model effectively predicts values of J_{SC} and V_{OC} close to those measured for devices of the respective molecules with the donor PM6; however, some deviations are observed. These deviations can be attributed to the inherent approximations in obtaining molecular properties from first-principles calculations. Ground-state and excited-state calculations and statistical sampling-based methods for obtaining various conformer configurations introduce certain errors in estimating device parameters. Nevertheless, we achieve a reasonable match with experimental values. However, it is noteworthy that calculating OPV efficiency requires additional information about the current−voltage characteristics, which we considered beyond the scope of this study.

Recently, NFAs with absorption ranges complementary to those of general donor polymers have been reported, maximizing the range of wavelength of absorption OPVs used to generate excitons.[73] However, our current model considers only the absorption of photons and the subsequent generation of excitons in the donor phase as there have been no reports specific to photoexcitation in the NFA systems considered here. Extending the model to include NFA absorption would involve simulating photon generation and diffusion in the acceptor domain. Subsequent steps, including CT formation and charge separation, are expected to follow a framework similar to that of the present model. Future extensions of our model could incorporate material-specific absorption data, enabling an exciton generation rate that more accurately reflects the combined donor and NFA absorption. This refinement would allow for closer alignment with experimental scenarios and more tailored predictions of J_{SC} across varied NFAs.

5. CONCLUSIONS

Using a multiscale approach, we devised a comprehensive computational framework for predicting the performance parameters of organic photovoltaic devices. Central to our method is a kMC algorithm that faithfully replicates the microscopic processes occurring within an operational OPV. By tracing the random walk of charge carriers based on the relative probabilities of each process, our model captures the intricate dynamics crucial to OPV functionality. Notably, our model stands out for its ability to predict both the J_{SC} and V_{OC} solely from molecular properties—a feat not achieved by previous models.

In our simulations, we simulate disordered donor−acceptor domains by varying the conformations of molecule pairs both within the bulk and at the donor−acceptor interface. This allows us to account for the morphological complexities inherent in OPV devices. Through extensive validation against measured devices, we demonstrate the model's remarkable accuracy in predicting J_{SC} and V_{OC} for various NFAs combined with the PM6 donor polymer. Furthermore, we leverage our model to evaluate the performance of new donor−acceptor combinations, identifying those that exhibit superior efficiency in conjunction with the donor PM6.

Our work extends beyond evaluation to design as we explore the OPV performance of newly designed acceptor molecules. By pinpointing molecular structures that yield high J_{SC} and V_{OC} values when paired with donor PM6, we offer valuable insights into molecular engineering in OPV applications. Our findings underscore the necessity of consolidating microscopic processes within a multiscale model to accurately assess OPV performance—an unachievable task by examining individual molecular properties alone. We show that the individual molecular properties are insufficient to select high-performing molecules, and a multiscale approach is necessary. Previously, no method has been reported that predicts both J_{SC} and V_{OC} of OPV based on molecular properties of the donor and acceptor, employing DFT-based methods and kMC simulations. Our multiscale model effectively combines molecular properties calculated from the first-principles calculation of molecular properties with the statistical treatment of thin-film properties to simulate charge carrier movement in a solar cell. To ensure that the method is computationally inexpensive, we have explored reasonable approximations to include accurate properties governing the microscopic processes in solar cells. Hence, the model described by us has a distinct advantage over previous models in evaluating and screening molecules for application in photovoltaics. In conclusion, our model represents a robust and computationally efficient tool for selecting and designing donor and acceptor molecules for OPV applications. Seamlessly integrating molecular properties with device-level characteristics offers a versatile platform for advancing the development of efficient organic solar cells.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.4c01029.

Molecular structures, tables of donor−acceptor combinations, and estimated V_{OC} from different approaches (PDF)

Compounds, conformers, and V_{AA}, V_{DA}, K_{CR}, K_{CS}, and K_{CT} values (xls)

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Notes

The authors declare no competing financial interest.

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